Chapter 11 exercises

State norms, scaling, and numerical assumptions. Give residuals or convergence arguments alongside computed vectors.

CH11-E1

Consider

$$A = \begin{bmatrix} \epsilon & 1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

with $0 < \epsilon \ll 1$.

1. Compare the first multipliers with and without partial pivoting.
2. Derive the exact symbolic solution by elimination.
3. Explain growth or cancellation as ϵ becomes small.
4. Discuss the determinant and what it does and does not establish about conditioning.

CH11-E2

1. Give leading flop counts for dense LU and triangular solves.
2. Explain factor reuse for 1000 right-hand sides.
3. Explain how initially sparse matrices can have dense factors.
4. Draw a star graph and compare the sparsity effects of center-first and leaves-first elimination.

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CH11-E3

Let

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 5 \end{bmatrix}.$$

1. Verify symmetry and positive definiteness through leading minors or energy.
2. Find the lower Cholesky factor satisfying $A = RR^T$.
3. Explain the leading arithmetic saving relative to general dense LU.
4. Analyze scaling by 10^{-8}, both for A alone and for A, b together.

CH11-E4

1. Prove $\|A\|_1$ equals the maximum absolute column sum.
2. Prove $\|A\|_\infty$ equals the maximum absolute row sum.
3. For $A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$, compute $\|A\|_1, \|A\|_\infty, \|A\|_F$.
4. Prove $\kappa_2(A) \geq 1$ for invertible square matrices.

CH11-E5

Let

$$A = 10^{-8} I_2, \qquad b = (1, 1)^T, \qquad \widehat{x} = (0, 0)^T.$$

1. Find the residual norm and true error.
2. Find relative forward error and normalized backward error.
3. Explain why an absolute residual alone does not certify quality.
4. State the relative right-hand-side perturbation bound.

CH11-E6

For

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \qquad b = (1, 2)^T.$$

1. Give the Jacobi iteration matrix and component updates.
2. Compute $\rho(G_J)$ and decide convergence.
3. Compute two iterates from $x_0 = 0$.
4. Distinguish componentwise monotonicity from contraction in a specified norm.

CH11-E7

For SPD

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = (3, 3)^T.$$

1. Write the quadratic energy and gradient.
2. Fix $x_0 = (0, 0)^T$. Derive the initial direction $p_0 = r_0 = b - Ax_0$ and first line-search step.
3. Starting from $x_0 = 0$, complete the exact-arithmetic solve.
4. Explain the limitations of the at-most-n-steps statement in floating point.

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CH11-E8

1. For invertible M, prove that $AM^{-1}y = b$, with $x = M^{-1}y$, is equivalent to $Ax = b$.
2. Discuss the mathematical effect and computational cost of choosing $M = A$.
3. Describe singular or poorly conditioned preconditioner failures.
4. Explain why eigenvalue moduli alone do not predict nonnormal GMRES residuals.
5. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $b = (1, 1)^T$, and $x_0 = 0$, compute two unrestarted GMRES Arnoldi steps. Give $v_1, v_2, \overline{H}_1, H_2$, the first reduced least-squares coefficient, x_1, r_1, and the second-step exact solution. Use the true residual to decide whether the zero remainder is a happy breakdown.

CH11-E9

Design a verification protocol:

1. Compare direct and iterative solutions.
2. Check residual, error estimate, and condition estimate.
3. Repeat with another ordering or initial guess.
4. Relate tolerance to scale rather than treating it as universal.

List at least five recorded quantities and explain two of them.

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CH11-E10

Combine the chapter's numerical methods.

Consider

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \qquad b = (1, 2)^T.$$

1. Verify that A is real SPD, and find its lower-triangular Cholesky factor R and solution x.
2. Find $\|A\|_2$, $\|A^{-1}\|_2$, $\kappa_2(A)$, and the CG energy-error bound from $x_0 = 0$.
3. For $\widehat{x} = (0.1, 0.6)^T$, find residual $r = b - A\widehat{x}$, error $e = x - \widehat{x}$, relative forward error, and normwise backward error.
4. Let

$$H = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}, \qquad x_c = (1, 2)^T, \quad b_c = Hx_c.$$

Verify Hermitian positive definiteness of H, and find its eigenvalues and $\kappa_2(H)$. With initial vector $x_0 = x_c$, what must complex CG do first?

5. Why is $r^T r$ not squared complex length? Distinguish incorrectly checking $H^T = H$ from replacing the operator by H^T. In this example, is the transposed operator still positive definite, and does it solve the same system?

CH11-E11

In exact arithmetic, take $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, b = (1, 0, 0)^T, x_0 = 0.$

1. Verify positive definiteness and compute all three CG steps, including $\alpha_k, \beta_k, p_k, x_{k+1}, r_{k+1}$.
2. Check all pairwise ordinary inner products of nonzero residuals and A-inner products of directions. Why need the directions not be ordinarily orthogonal?
3. Derive energy minimization from $r_k \perp \mathcal{K}_k$. Explain the short recurrence and complete the at-most-n-step dimension argument.

CH11-E12

1. Let $A = \operatorname{diag}(1, \varepsilon)$, $0 < \varepsilon < 1$, and $E = \operatorname{diag}(0, -\delta)$, $0 < \delta < \varepsilon$. Find the relative 2-norm change of the inverse and determine when the exact perturbation bound is attained.
2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $b = (1, 1, 0)^T$. A full orthogonal matrix has columns

$$q_1 = (1, 0, 1)^T/\sqrt{2}, \quad q_2 = (-1, 2, 1)^T/\sqrt{6}, \quad q_3 = (-1, -1, 1)^T/\sqrt{3}.$$

Compute $Q^T A$, $Q^T b$, and solve the correctly sized triangular least-squares system. Find the residual squared and distinguish the full-Q and reduced-Q_1 right-hand sides.

CH11-E13

For A=diag(1,100), b=(1,1/10)ᵀ and x_0=0, calculate the first exact CG step. Compare squared residual norms and energy errors $F(x_k)-F(x^*)$ before and after this step. Does a larger residual contradict CG energy minimization? Separately, for A=[[4,1],[1,3]], verify $p_0^T A p_1 = 0$ but $p_0^T p_1 \neq 0$ using $p_0=(1,2)^T$, $p_1=(-7/16,3/8)^T$, and distinguish this from $r_0^T r_1 = 0$.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 11.

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